Richer than they feel? Thirty years of purchasing power in Brazil

Klaus Colletti Oehling[†]

Preliminary draft. This version: October 2026.

Abstract

A common perception in Brazil holds that purchasing power has been falling. Using household survey microdata for 1995–2025, I show that real household per capita income rose at every percentile, by 191% at the 10th, 143% at the median and 66% at the 90th, under the general consumer price index and under each of its nine expenditure groups. Individual mobility, heterogeneous baskets and debt service can hide individual losses, but are unlikely to explain the perception on their own. Structural-break and Markov-switching estimates show that real income stagnated across most of the distribution in 1995–2003 and 2014–2022, sixteen of the thirty years, with the largest swings at the bottom. Drawing on the literature on perceived inflation, reference-dependent preferences and economic news, I argue that these periods of loss, together with biases in how households perceive prices and income and a possible negativity bias in the news, sustain the perception.

Keywords: purchasing power; perceived inflation; income distribution; Brazil.
JEL Classification: C22; D31; D84; D91; E31; O54.

1 Introduction

In July 1994, the Plano Real ended more than a decade of chronic high inflation in Brazil (Giambiagi et al., 2016). Three decades later, a common perception is that money buys less and less, and that Brazilians have been losing purchasing power. Polls measure only its twelve-month version: in the Genial/Quaest series, the share of respondents who said their purchasing power was lower than a year before was 30% in June 2023 and 72% in November 2025 (Genial/Quaest, 2024, 2025). This paper asks whether purchasing power in Brazil has in fact fallen since the Real and, if it has not, why the perception of loss is so common.

[†]klausco@al.insper.edu.br

The first part of the paper establishes the facts. With household per capita income from the National Household Sample Survey (PNAD) for 1995–2011 and from the Continuous PNAD for 2012–2025, I measure real income growth at every percentile with growth incidence curves (Ravallion and Chen, 2003). Because a single price index can misrepresent the cost of living of households with different baskets (Prais, 1959; Jaravel, 2021), incomes are deflated by the general Extended National Consumer Price Index (IPCA) and by each of its nine expenditure groups.

Real household per capita income grew at every percentile between 1995 and 2025: by 191% at the 10th percentile, 143% at the median and 66% at the 90th, or 3.6%, 3.0% and 1.7% a year. Growth is positive under every deflator, and the Gini coefficient fell from 0.60 to 0.51. The gains, however, were concentrated in time. Dating periods of gain and loss at every percentile with structural-break (Bai and Perron, 2003) and Markov-switching (Hamilton, 1989) methods, I find that real income stagnated across most of the distribution in 1995–2003 and again in 2014–2022, sixteen of the thirty years, and that nearly all of the gain accrued in 2003–2014 and 2022–2025.

The second part asks why perception and measurement diverge. Growth incidence curves compare percentiles, not people: the 10th percentile of 1995 and that of 2025 are different individuals, so many people can lose while every percentile gains. Together with baskets that differ from the IPCA's and the cost of debt, this can produce real individual losses without any misperception. These features are unlikely to explain the perception on their own, and the rest of the gap lies in how purchasing power is perceived. The periods of stagnation supply the raw material, and mechanisms documented in the literature on how households perceive prices, income and economic news reinforce it. Table 1 lists the mechanisms that, based on this literature and on the Brazilian data, I argue best account for the perception: losses that the distributional comparison may hide, which Section 4.2 delineates with the data, and ways in which perception departs from experience, which Section 5 discusses.

The paper is descriptive: it establishes the facts that any explanation must match and confronts each candidate mechanism with the Brazilian evidence, without estimating their contributions. It offers a thirty-year view of the whole income distribution that is robust to the choice of deflator, a dating of the periods of gain and loss at every percentile, and a separation between what the structure of the data can explain and what requires

misperception. Section 2 reviews the literature, Section 3 describes the methods and data, Section 4 presents the results, Section 5 discusses the perception and Section 6 concludes.

Table 1: Mechanisms behind the perception of lost purchasing power examined in this paper

Mechanism	Description
Losses hidden by the distribution (Section 4.2)	
Individual mobility	Percentiles are not people: many individuals can lose while every percentile gains
Basket heterogeneity	Households whose budgets differ from the IPCA basket face a different inflation rate
Cost of money	Debt service, excluded from price indices, reduces the income available for consumption
Perception (Section 5)	
Periods of loss	Long periods of stagnation and loss weigh more than gains of the same size and shape beliefs that outlast them
Purchase frequency	The prices of frequently purchased goods dominate perceived inflation
Loss aversion over prices	Price increases weigh more than price decreases of the same size
Nominal memory	Current prices are judged against remembered nominal prices
Attribution	Income gains are credited to merit, price increases to inflation
Aspirations	Aspirations rise with income and absorb part of each gain
Relative position	A loss of relative standing is felt despite absolute gains
Short horizon	Polls ask about the last year, and news tracks recent changes rather than levels
Negative news	Bad economic news may receive more coverage and attention than good news
Partisanship	Assessments of the economy track political allegiance
Social transmission	Beliefs spread through social networks, amplifying the mechanisms above

2 Related literature

2.1 Measuring who gains from growth

Growth incidence curves (GICs) plot the growth rate of income at each quantile of the distribution between two dates (Ravallion and Chen, 2003). The standard GIC is anonymous: it compares quantiles of two cross-sections, which are generally occupied by different people. Non-anonymous GICs follow the same individuals and plot their income growth against their initial position (Grimm, 2007). The two can diverge: in the panels studied by Jenkins and Van Kerm (2006) and Grimm (2007), individual income growth

favored the initially poor more than anonymous curves suggest, even where inequality rose. This paper uses anonymous curves; Section 4.2 discusses what they imply for individuals.

2.2 Whose price index?

A standard consumer price index weights households by their expenditure and therefore tracks the basket of a relatively well-off household (Prais, 1959), and inflation has tended to be lower for higher-income households (Jaravel, 2021). The IPCA covers urban families with income between 1 and 40 minimum wages (IBGE, 2026), with weights from the household budget survey (POF). Budget shares differ sharply across incomes: families earning up to two minimum wages spend 61.2% of their budget on food and housing, against 30.2% among those earning more than 25 (IBGE, 2019). Instead of class-specific indices, this paper deflates incomes by the general IPCA and by each of its nine groups, which bounds the purchasing power of households whose budgets concentrate in any group.

2.3 Perceived and measured purchasing power

Comparisons between perceived and measured purchasing power cover short horizons. In the euro area, real disposable income rose 3.8% between mid-2022 and mid-2024, yet only 37% of households said their real income had increased or stayed the same, although more than half had experienced positive real income growth (Baumann et al., 2025). Linking survey answers to the same households' verified purchases, Hacıoğlu Hoke et al. (2025) find that 43% of US consumers reported being worse off than in 2019, while most had increased their real spending. In panels for Peru and Russia, Graham and Pettinato (2002) identify "frustrated achievers", whose incomes rose but who reported being worse off. This paper adds a thirty-year benchmark for the whole income distribution of an emerging economy with a history of high inflation.

2.4 The income distribution in Brazil

Studies based on the PNAD document falling inequality after the Real. The Gini coefficient fell from 0.63 in 1989 to 0.56 in 2004, and its fall after 1993 is attributed to lower inflation, declining returns to education, rural–urban convergence and better-

targeted social transfers (Ferreira et al., 2008). Conditional cash transfers account for about a fifth of the fall in inequality between the mid-1990s and the mid-2000s (Soares et al., 2009) and Bolsa Família for about a tenth of the fall between 2001 and 2015 (Souza et al., 2019), while the 128% real increase in the minimum wage between 1996 and 2018 explains about 45% of the decline in earnings inequality (Engbom and Moser, 2022). Tax records show that household surveys understate incomes at the top (Medeiros et al., 2015): combining national accounts, surveys and tax data, Morgan (2017) finds that the top 10% captured about 61% of income growth between 2001 and 2015. The results of this paper for the upper percentiles rely on survey data only.

3 Methodology

3.1 Statistical tools

For each year t, individuals are ranked by household per capita income and the weighted empirical distribution function F_t is computed from the cumulative sum of the sampling weights. The quantile function $y_t(p) = F_t^{-1}(p)$, the income at percentile p, is obtained by linear interpolation on a grid of 10,001 equally spaced points between 0 and 1. Nominal incomes are converted into constant prices with a deflator k, the general IPCA or one of its nine groups, measured in September, the reference month of the annual PNAD:

$$y_t^k(p) = y_t(p)\,\frac{P_{2025}^k}{P_t^k}. \tag{1}$$

The growth incidence curve (Ravallion and Chen, 2003),

$$g^k(p) = \left[\frac{y_{2025}^k(p)}{y_{t_0}^k(p)}\right]^{1/(2025-t_0)} - 1, \tag{2}$$

is the annualized real growth of income at percentile p, with $t_0 = 1995$, except for transportation, communication and education, for which $t_0 = 2001$. The nominal curve is computed from $y_t(p)$ without deflation. Deflating by a group instead of the general index shifts the whole curve by approximately the difference between the two inflation rates. Where income is zero in the initial or final year, which occurs only below the 2nd percentile, growth is not defined. The set of deflators works as a stress test: if real income

grew at a percentile under every deflator, a reported loss at that percentile cannot be attributed to the choice of price index among the IPCA's components.

The same growth rate computed between consecutive surveys $s < t$, $\left[y_t^k(p)/y_s^k(p)\right]^{1/(t-s)} - 1$, follows the timing of the changes, and its variance across surveys measures the volatility of real growth at each percentile. To gauge which part of the distribution the IPCA represents, I locate in each year the percentiles whose household per capita income is closest to one and to forty times the December minimum wage; since the IPCA's income range refers to total family income, this approximation understates its coverage at the bottom. Inequality is measured by the Gini coefficient,

$$G_t = 1 - 2\int_0^1 L_t(p)\,dp, \tag{3}$$

where the Lorenz curve $L_t(p)$ is the share of total income received by the poorest fraction p of the population, and by the income shares of the bottom 50%, $L_t(0.5)$, the top 10%, $1 - L_t(0.9)$, and the top 1%, $1 - L_t(0.99)$.

To date the periods of gain and loss, I treat the real growth of each percentile between consecutive surveys, $g_s(p)$, deflated by the general IPCA, as a series of 26 observations from 1996 to 2025 whose mean shifts between regimes,

$$g_s(p) = \mu_{r(s)}(p) + \varepsilon_s(p), \tag{4}$$

where $r(s)$ is the regime of survey s. Two methods identify the regimes. The structural-break method of Bai and Perron (1998, 2003) estimates, for each number of breaks $m \leq 5$, the break dates that minimize the sum of squared residuals, with at least three surveys per regime, selects m by the Bayesian information criterion and gives each break date a 90% confidence interval (Bai, 1997). The Markov-switching model of Hamilton (1989) lets two regimes, of low and high growth and with a common variance, alternate over time, and assigns each survey a smoothed probability of the low-growth regime; it is estimated from 20 random starting values, keeping the highest likelihood. The first method dates one-off breaks; the second allows a regime to return.

A Bai–Perron segment is classified as low growth if its mean is closer to the Markov low-regime mean than to the high-regime mean, and a survey as Markov-low if its probability of the low regime exceeds one half. At each percentile, a survey is then classified

as low growth if both methods say so, as high growth if both say so, and as undetermined otherwise. Consecutive surveys with the same classification form a period, whose cumulative growth is annualized over its length,

$$\bar{g}(p) = \left[\frac{y_{t_1}(p)}{y_{t_0}(p)}\right]^{1/(t_1 - t_0)} - 1, \tag{5}$$

where t_0 is the survey preceding the period and t_1 its last survey. At the very bottom, surveys with undefined growth are left out, and percentiles with fewer than 15 valid surveys are not classified. With 26 observations per series, regimes are estimated with limited precision; the agreement of two methods with different assumptions is the main safeguard. The appendix reports the classification for every percentile (Figure A4) and both methods at five percentiles (Figure A5).

3.2 Data

The analysis uses the microdata of the National Household Sample Survey (PNAD) of the Brazilian Institute of Geography and Statistics (IBGE): the annual PNAD for 1995–2011 and, for 2012–2025, the first-visit annual microdata of its successor, the Continuous PNAD (PNAD Contínua). Household per capita income (RDPC) is IBGE's variable VD5008 in the PNAD Contínua and V4742 in the PNAD for 2004–2011. The 1995–2003 editions lack a ready-made measure, so RDPC is computed as total household income (V4614) divided by the number of household members (V0105). Unlike IBGE, this counts live-in domestic workers, their relatives and boarders as members, a small difference that is neglected. The sampling weights are V4729 for 1995–2011 and V1032 for 2012–2025. The unit of analysis is the person: each individual is assigned the per capita income of their household and weighted by their own sampling weight.

The sample starts in 1995: there was no PNAD in 1994, and earlier surveys precede the Real. There is no PNAD for the census years 2000 and 2010, and first-visit microdata are not available for 2020 and 2021. Two changes in survey design matter. From 2004, the PNAD covers the rural areas of the North region, previously excluded, and in 2012 the PNAD Contínua replaces the annual PNAD. Time-series figures do not connect observations across these breaks, marked by dotted vertical lines, and the growth rates into 2004 and 2012, which combine income growth with the change in design, are marked

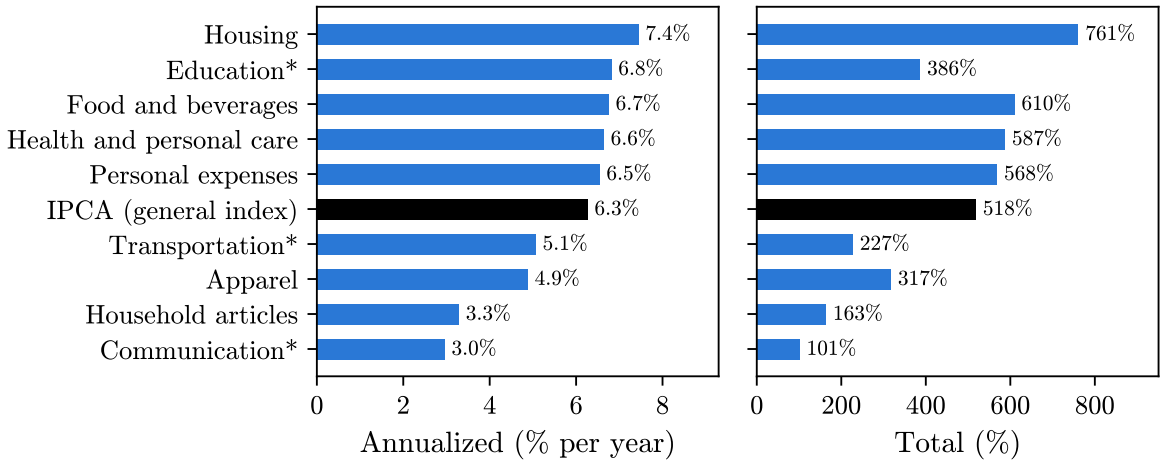
with an asterisk.

Prices are measured by the IPCA and its nine expenditure groups: food and beverages, housing, household articles, apparel, transportation, communication, health and personal care, personal expenses, and education. The series for transportation, communication and education start in August 1999, so, for these three groups, growth is measured from 2001, the first survey year with a full year of their prices. The price series and the nominal minimum wage come from the Central Bank of Brazil's time-series system (SGS). All figures are the author's calculations from these sources.

4 Results

4.1 Has purchasing power fallen?

Prices rose 518% between September 1995 and September 2025, or 6.3% a year (Figure 1). Housing (7.4% a year), food and beverages (6.7%), health and personal care (6.6%) and personal expenses (6.5%) outpaced the general index, while apparel (4.9%) and household articles (3.3%) lagged behind it. Between 2001 and 2025, education rose 6.8% a year, transportation 5.1% and communication 3.0%. Relative prices moved against food and housing and in favor of household articles and apparel, a dispersion that is one reason to test the results against each group and that reappears in Section 5.2.



* Sep 2001 to Sep 2025 (series start in Aug 1999); all others Sep 1995 to Sep 2025.

Figure 1: Accumulated inflation of the IPCA and its groups

Another reason is that the general IPCA tracks the prices of urban, middle-income

households. Under the per capita approximation, one minimum wage, the lower bound of its target population, falls between the 48th and the 60th percentile of household per capita income throughout the period, while forty minimum wages exclude only the top 0.1% (Figure A1).

Under the general IPCA, real income grew at every percentile (Figure 2): 3.6% a year at the 10th percentile, 3.0% at the median, 1.7% at the 90th and 1.3% at the 99th, or 191%, 143% and 66% in total at the 10th, 50th and 90th percentiles. Growth was faster for the poor than for the rich. Below the 5th percentile, where incomes approach zero, growth rates are large and imprecise.

Changing the deflator shifts the curve but never changes its sign. Groups whose prices rose faster than the general index lower it, and housing lowers it most: under housing, growth falls to 2.5% a year at the 10th percentile, 1.9% at the median, 0.6% at the 90th and 0.2% at the 99th. Groups whose prices rose slowly raise it; under household articles, growth exceeds 4% a year at every percentile.

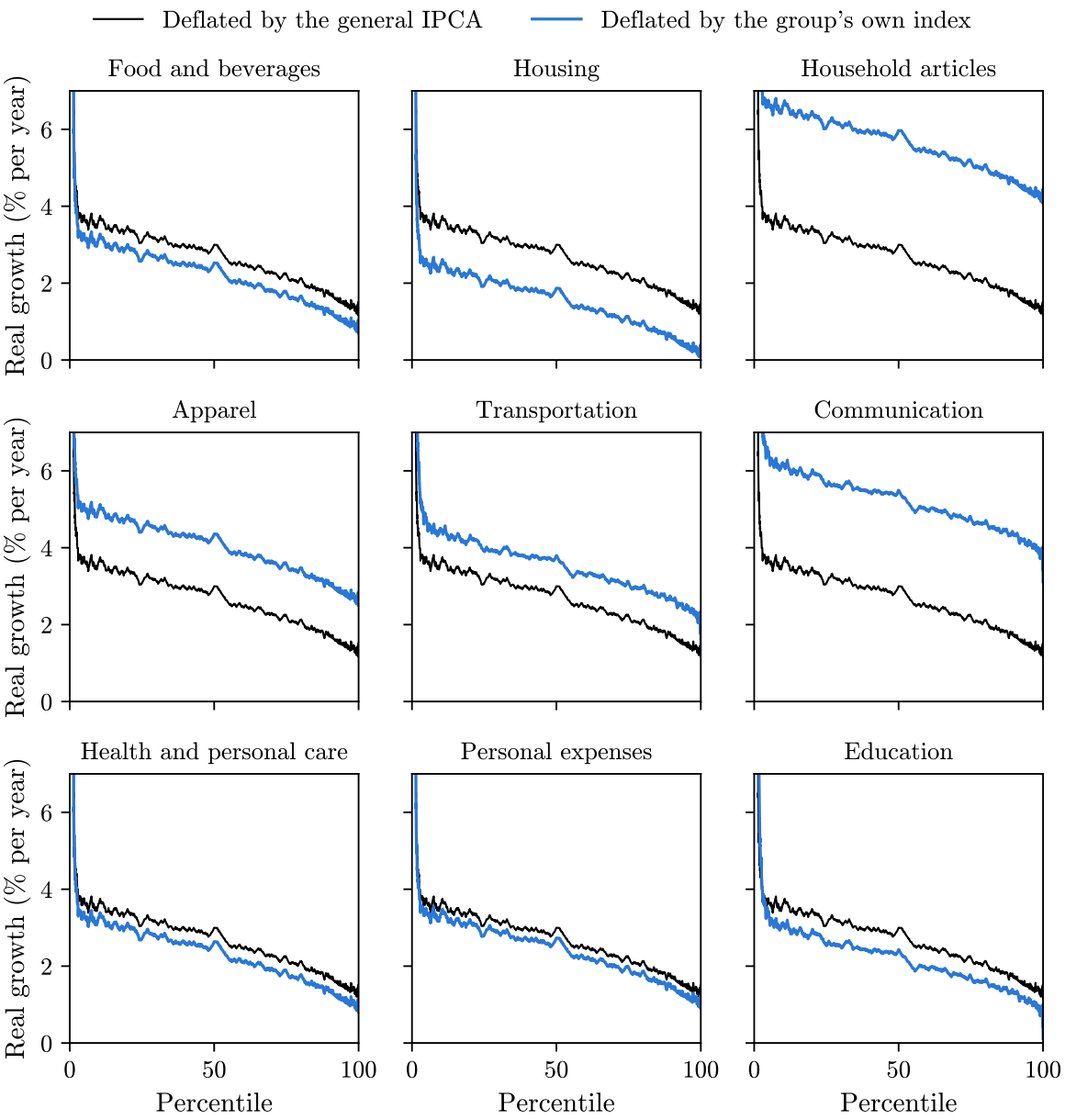


Figure 2: Growth incidence curves by IPCA group, annualized

Nominal income grew between 7.6% and 10.1% a year across the distribution, and the whole range of real growth rates obtained with the nine groups lies above zero at every percentile (Figure 3): no deflator among the IPCA's components produces a loss of purchasing power for any part of the distribution.

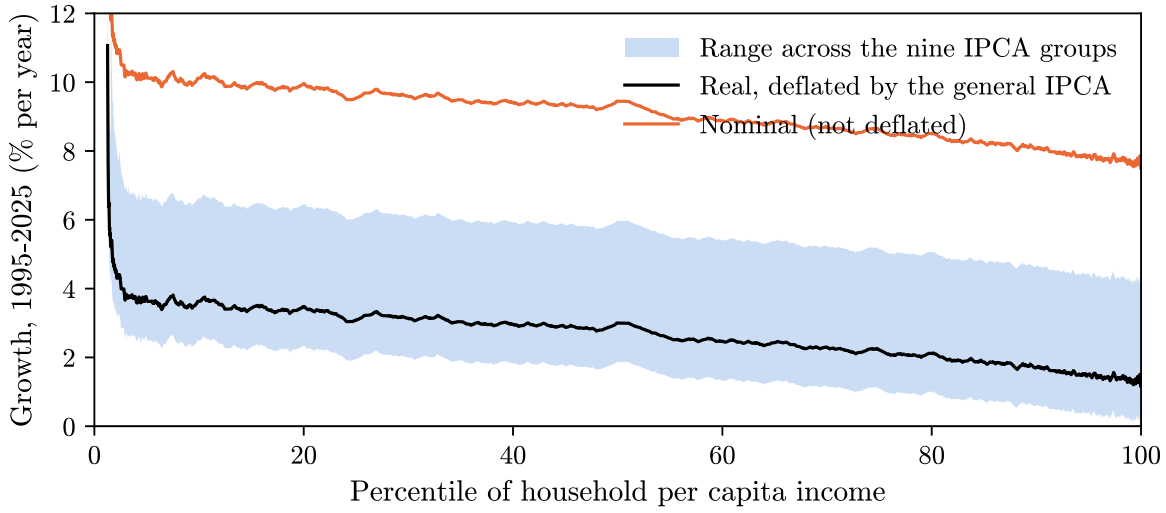


Figure 3: Nominal and real growth incidence curves, 1995–2025

Two measurement issues do not alter this result. Household surveys understate top incomes (Medeiros et al., 2015), so the upper tail is measured with error, but this cannot reverse the result for the rest of the distribution. And the comparison spans two changes in survey design: even if the entire changes into 2004 and into 2012 were attributed to the redesigns, real income would have grown by more than 40% at every percentile between the 5th and the 95th.

The thirty-year comparison hides an uneven path (Figure 4). Between 1995 and 2003, real income stagnated, and at the 90th percentile it fell 8%. Between 2004 and 2011, it rose 63% at the 10th percentile, 57% at the median and 27% at the 90th, and it kept rising until 2014. The 2015–2016 recession reversed part of these gains: real income fell 16% at the 10th percentile between 2014 and 2018 and 8% at the median between 2014 and 2016, and the 10th percentile regained its 2014 level only in 2022. Between 2022 and 2025, real income rose 29% at the 10th percentile, 26% at the median and 17% at the 90th. The real minimum wage followed the same timing: it roughly tripled between 1995 and 2025, from about R$ 500 to R$ 1,518 at December 2025 prices, with most of the gain between 2003 and 2014 and little change between 2014 and 2022 (Figure A2).

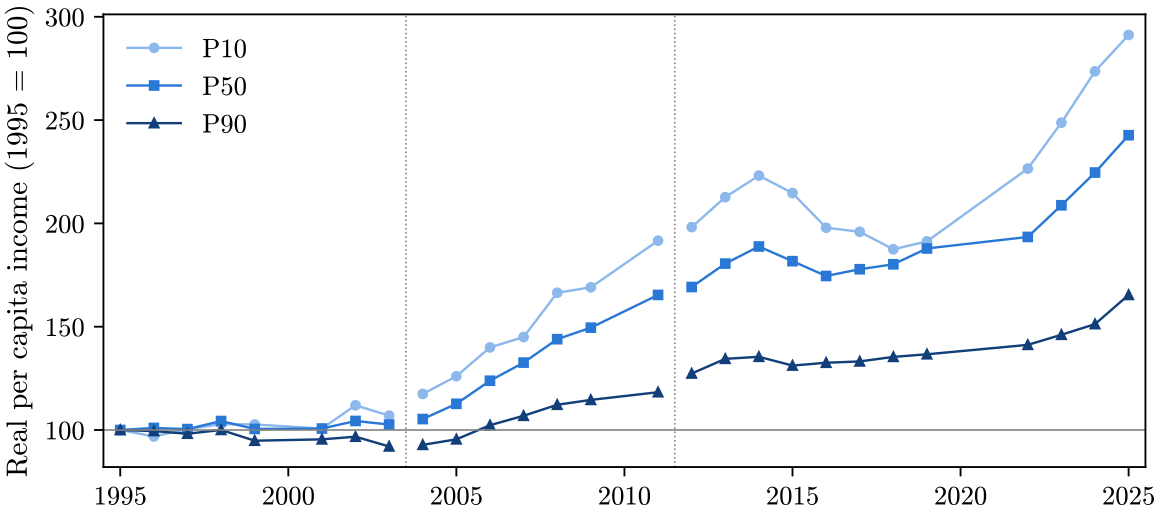


Figure 4: Real household per capita income at the 10th, 50th and 90th percentiles

Inequality fell (Figure 5): the Gini coefficient declined from 0.596 in 1995 to 0.511 in 2025, after stalling before 2003 and rising in 2015–2018.

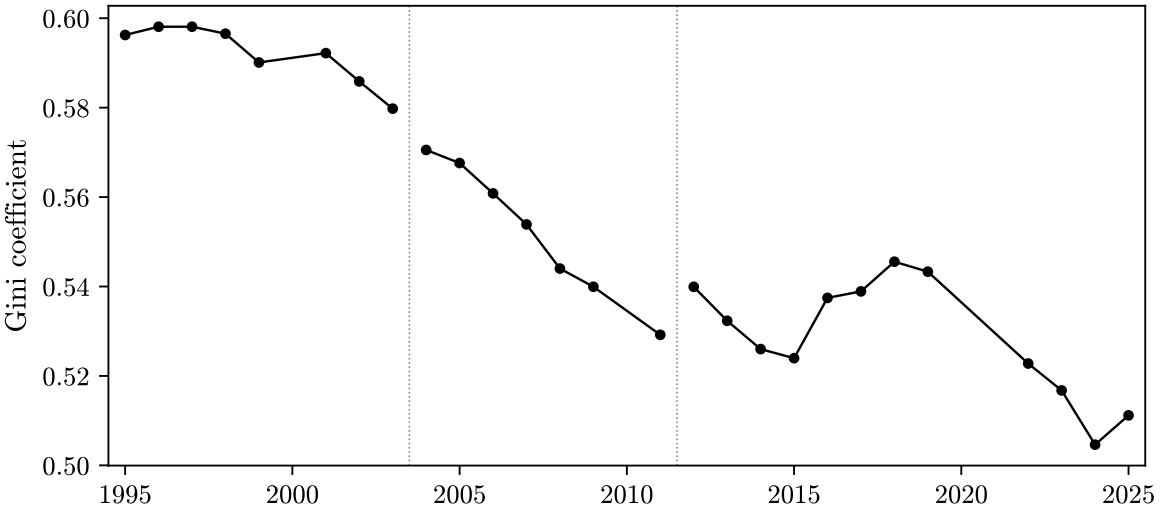


Figure 5: Gini coefficient of household per capita income

The share of income received by the poorer half of the population rose from 12.5% to 17.5%, while the share of the top 10% fell from 47.3% to 40.3% and that of the top 1% from 13.4% to 11.0% (Figure A3).

4.2 Losses hidden by the distribution

The comparison above concerns the purchasing power of the distribution; polls ask about the purchasing power of individuals. Three features of the data can make the two

diverge without any misperception: the comparison follows percentiles rather than people, price indices follow an average basket, and income measures ignore the cost of credit.

Let $y_{i,t}$ denote the real income of individual i in year t. Section 4.1 establishes that $y_{2025}(p) > y_{1995}(p)$ at every percentile p, so that the 2025 distribution first-order stochastically dominates that of 1995. It does not establish that $y_{i,2025} > y_{i,1995}$ for most individuals. Cross-sections identify the distribution of income in each year, not the joint distribution that links each person's incomes at the two dates. The two distributions fix the average of individual changes, which equals the change in mean income and is therefore positive: in money terms, gains exceed losses. They do not fix how many people gained; the share of losers can lie anywhere between zero, if everyone keeps their rank, and nearly one.

An example shows the second extreme. In a population of 100, let the individual of rank n earn $100n$ in the first period and $100n + 1$ in the second, so that every percentile gains one unit. Suppose that the poorest individual moves to the top and everyone else moves down one rank. The former gains 9,901 and each of the other 99 loses 99: every percentile is richer and mean income has risen by one unit, yet 99% of the population is poorer. Polls count people, whereas income aggregates add money; a poll in this economy would record a near-unanimous loss, and it would be right. For such losses to outweigh the aggregate gain in welfare, the mechanism needs a single behavioral assumption: that the value of income is concave, or that losses weigh more than gains of the same size (Kahneman and Tversky, 1979), as income losses do in British and German panels (Boyce et al., 2013).

Whether mobility matters depends on the horizon. Over thirty years, it is unlikely to produce widespread losses: real income rose by 66% or more at every percentile from the 5th to the 90th, so an individual lost purchasing power over the period only by falling far down the distribution, and in panel data individual growth tends to favor the initially poor (Jenkins and Van Kerm, 2006; Grimm, 2007). Over the twelve months that polls ask about, mobility is central. Brazilian household incomes are volatile: across PNAD Contínua interviews twelve months apart, the share of people poor in at least one of the two interviews exceeds the cross-sectional poverty rate by more than half at the lowest poverty line, and more than 70% of those ever poor are poor in only one of them (Souza et al., 2026). Many Brazilians may therefore lose real income in a year in which every

percentile gains, and the question the polls ask, whether one's own purchasing power fell, is a non-anonymous one.

Inflation also differs across households (Section 2). Households whose budgets differ from the IPCA basket, such as renters or families that spend most of their income on food, face a different rate. The stress test of Section 4.1 bounds this channel: no IPCA group, taken as a household's entire budget, turns real growth negative at any percentile. Basket heterogeneity can make gains smaller for households concentrated in housing; it is unlikely, by itself, to turn them into losses.

Consumer price indices also exclude interest payments, although, for households that borrow, higher interest rates raise the cost of living: in the United States, inflation measures that include borrowing costs explain about three quarters of the 2023 gap between consumer sentiment and fundamentals (Bolhuis et al., 2024). Income committed to debt service is not available for consumption, so this channel separates measured from experienced purchasing power without any misperception.

These features can make a report of lost purchasing power accurate for many Brazilians, but they are unlikely to explain the perception on their own: where surveys have matched perceptions to the same households' measured incomes or spending, many households whose real income or spending rose reported a loss (Baumann et al., 2025; Hacıoğlu Hoke et al., 2025; Graham and Pettinato, 2002). The rest of the gap lies in how purchasing power is perceived, the subject of the next section.

5 Discussion

Household perceptions and expectations of inflation are, on average, biased upward, dispersed and shaped by personal shopping experience (Weber et al., 2022), and the inflation expectations of Brazilian consumers exceed realized inflation, more so at lower incomes (Teixeira et al., 2017). In Brazil, the clearest sign of a gap between reported and measured purchasing power is recent: in November 2025, 72% of respondents said their purchasing power was lower than a year before (Genial/Quaest, 2025), while measured real income rose 6% at the 10th percentile and 8% at the median between the 2024 and 2025 surveys. Part of this gap may reflect individual losses in a year in which every percentile gained (Section 4.2); the rest requires a gap between experience and perception. I begin

with the periods of loss, in which the data supply the raw material, and then group the remaining mechanisms into what households observe, how they evaluate it, and the information environment in which their beliefs form. Biased observation can, in principle, be corrected with better information; biased evaluation cannot. The implications drawn for Brazil are hypotheses, not estimates.

5.1 Periods of loss

Figure 6 extends the path of Figure 4 to every percentile: real losses concentrate in 1999, 2003 and 2015–2017, and gains in 2005–2014 and 2023–2025.

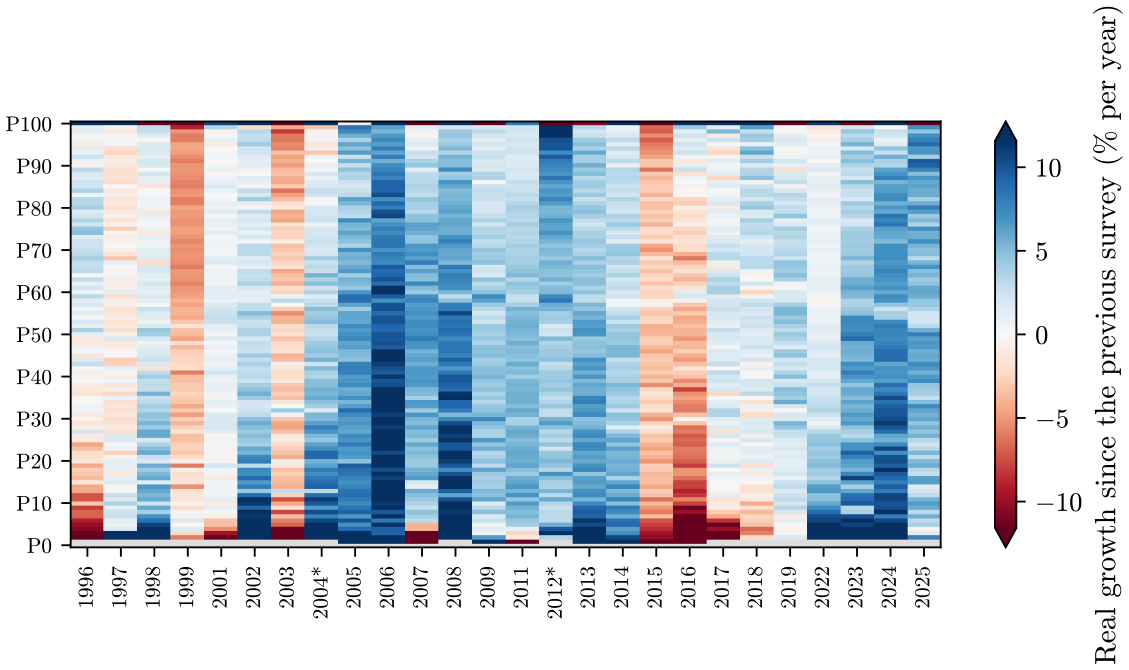


Figure 6: Real growth between consecutive surveys, by percentile

Note: Annualized real growth since the previous survey, deflated by the general IPCA. * The changes into 2004 and 2012 combine income growth with changes in survey design (Section 3.2).

The regime analysis turns these swings into periods. Between the 5th and the 95th percentiles, both methods place 79% to 92% of the percentiles in the low-growth regime in every survey from 1996 to 2003 and 76% to 79% in 2015–2017, and 95% to 98% in the high-growth regime in 2005–2013 and 2023–2025 (Figure A4). The exit from the second low-growth period was uneven: the bottom fifth of the distribution returned to high growth in 2019–2022, while a band roughly between the 30th and the 65th percentiles remained in the low-growth regime until 2022. Figure 7 measures the growth of each period. In the high-growth regime of 2004–2014, annualized real growth ranged from

about 4% a year at the top of the distribution to about 7% at the bottom; in the low-growth regime that followed, it was −4.3% a year at the 10th percentile and close to zero further up.

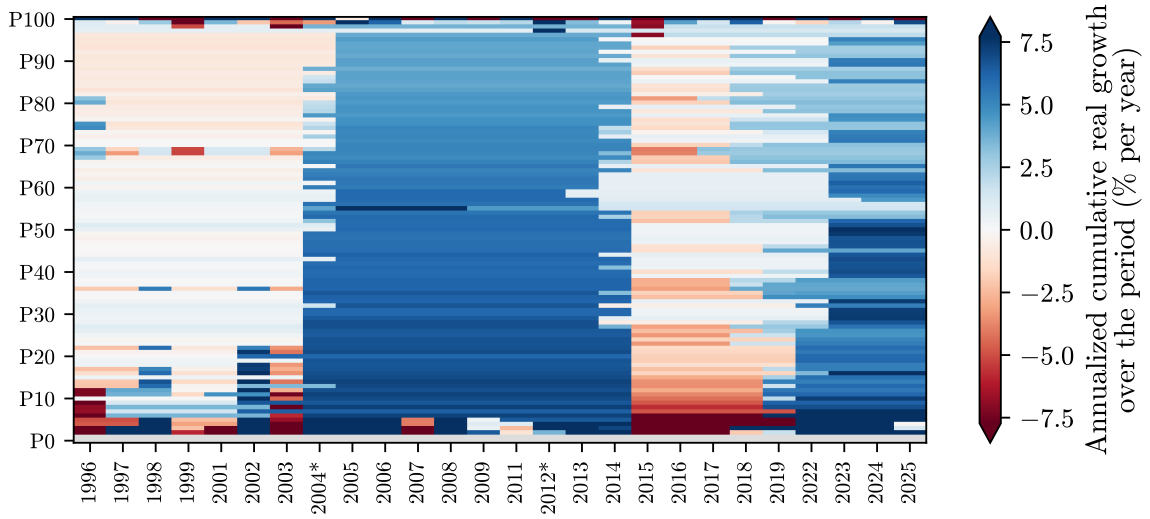


Figure 7: Real growth within each consensus period, by percentile

Note: Each cell shows the cumulative real growth of the period it belongs to in Figure A4, annualized by the period's length in years (Equation 5). * As in Figure 6.

Taken together, the two low-growth periods cover sixteen of the thirty years: at every percentile between the 5th and the 95th, real income in 2003 stood within 10% of its 1995 level, and in 2022 within 10% of its 2014 level. Nearly all of the thirty-year gain accrued in the remaining fourteen years: between 2003 and 2014, real income rose 109% at the 10th percentile, 84% at the median and 47% at the 90th, and between 2022 and 2025, 29%, 26% and 17%.

The bottom of the distribution gained the most, but along the most volatile path (Figure 8): the standard deviation of real growth between consecutive surveys is 5.8 percentage points at the 10th percentile, against 4.0 at the median and 3.5 at the 90th, and it rises steeply below the 10th percentile. Over windows of several years, a report of lost or stagnant purchasing power therefore describes the data. Reports of loss in years of measured gain, such as 2025, can stem from these periods only if they weigh more than their length warrants, and the literature gives three reasons why they may. First, losses weigh more than gains of the same size (Section 4.2), so a volatile path is valued below a smooth one with the same total gain, most at the bottom of the distribution, where the swings were largest.

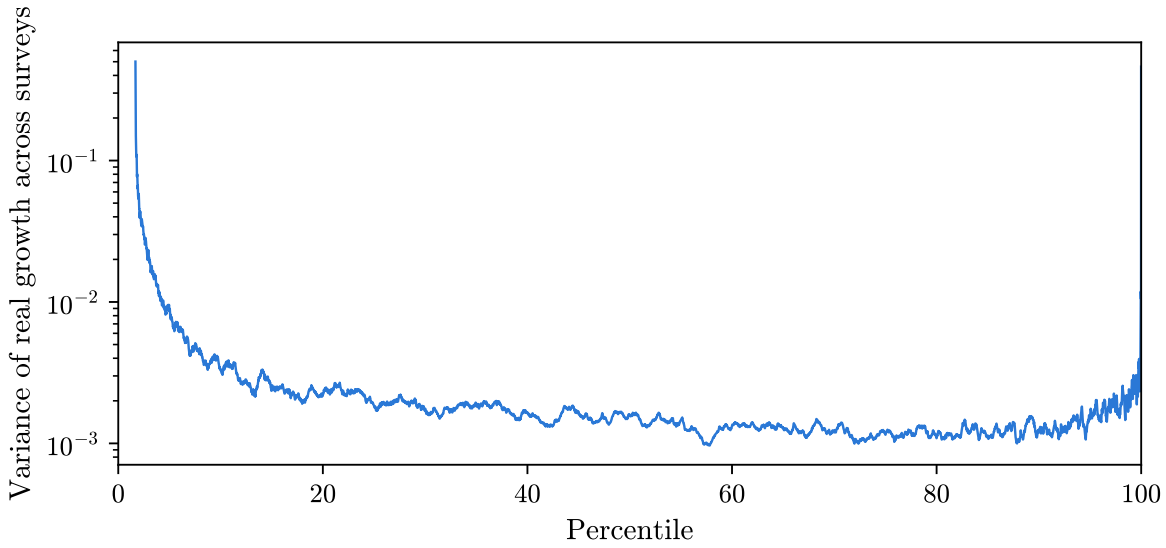


Figure 8: Volatility of real growth by percentile

Note: Variance across surveys of the annualized real growth since the previous survey (general IPCA, decimal units, log scale), including the changes into 2004 and 2012. Percentiles with zero income in some survey are omitted.

Second, a recovery need not register as a gain. When the reference point is the previous peak (Duesenberry, 1949) or recent expectations (Kőszegi and Rabin, 2006), a recovery registers as the reversal of a loss: from 2014 to 2022, the 10th percentile lived below a level it had already reached. Across Latin America, social discontent in the 2010s tracked the deceleration relative to the 2000s boom rather than the level of income (Albina et al., 2025).

Third, beliefs formed in bad periods outlast them. Individuals weight the economic conditions they have lived through when forming expectations (Malmendier and Nagel, 2016), extrapolate from personal experiences, such as a spell of unemployment, to aggregate outcomes (Kuchler and Zafar, 2019), and absorb new information only gradually (Carroll, 2003). The 2022–2025 recovery followed eight years of stagnation, and beliefs formed in those years may outlast it. The losses were real; the weight they retain in beliefs after measured incomes resumed growing is a bias.

5.2 What households observe

Households do not observe the consumer price index; they observe the prices of what they buy, weighted by how often they buy it. Jonung (1981) attributed the higher inflation perceived by Swedish women to their greater exposure to food prices. In laboratory

experiments, Georganas et al. (2014) find that perceived inflation places a weight of about 44% on purchase frequency, where the correct weight is zero, and D'Acunto et al. (2021) confirm with scanner data that the price changes of a household's own purchases, weighted by frequency, predict its inflation expectations better than the same changes weighted by expenditure. Brachinger (2008) formalizes these findings in an index of perceived inflation that weights goods by purchase frequency and, following loss aversion (Kahneman and Tversky, 1979), gives price increases more weight than decreases of the same size.

In Brazil, relative prices moved against frequent purchases. Food and housing, which include groceries and monthly bills, rose faster than the general index, while household articles and apparel rose much more slowly, so that annual inflation ranged from 3.3% for household articles to 7.4% for housing (Figure 1). A perception that weights goods by purchase frequency and penalizes price increases would therefore exceed the IPCA, and incomes deflated by it would appear to have grown less. The mechanism is bounded by Section 4.1: deflated by housing, the fastest-rising group, real income still grew at every percentile. Purchase frequency and loss aversion can shrink perceived gains; on their own, they are unlikely to turn them into losses.

Households also judge today's prices against remembered ones. People represent transactions in nominal as well as real terms, and the nominal representation biases judgment (Shafir et al., 1997); in the United States, about a third of the post-2021 gap between consumer sentiment and fundamentals has been attributed to the cumulative rise in the price level rather than to the inflation rate (Cummings et al., 2024). The Brazilian price level rose more than sixfold between 1995 and 2025. Against remembered nominal prices, current prices look high regardless of what happened to incomes, and more so the longer the horizon of the comparison.

5.3 How households evaluate

Purchasing power is the ratio of income to prices, and households track the two sides differently. Shiller (1997) found among American, German and Brazilian respondents the belief that inflation lowers living standards because incomes do not keep pace with prices. Individuals attribute their own raises to job performance rather than to inflation, so nominal income growth that compensates for inflation is not perceived as compensation (Stantcheva, 2024), and even when wages keep pace, workers experience the process as a

loss, because keeping pace requires costly conflict, such as asking for raises or changing jobs (Guerreiro et al., 2026). In Brazil, minimum wage increases and transfers account for a substantial part of the fall in inequality, that is, of the faster growth at the bottom (Engbom and Moser, 2022; Soares et al., 2009; Souza et al., 2019). These are discrete, institutional adjustments, which households may attribute to government decisions or personal circumstances rather than read as compensation for prices, which rise continuously and visibly at every purchase. If gains are credited elsewhere and losses are charged to prices, the evaluation of purchasing power is biased downward even when every price is perceived correctly.

Well-being depends on income relative to aspirations, and aspirations rise with income, absorbing part of each gain, a mechanism behind the Easterlin paradox (Easterlin, 1995). The income households report as necessary to get by rises with their own income (Goedhart et al., 1977), and Brazil is no exception: in the 2017–2018 POF, the income reported as necessary rose from R$ 470 a month in the lowest income group, almost twice its disposable income of R$ 245, to R$ 4,001 in the highest (IBGE, 2020). Unlike the reference points of Section 5.1, anchored in incomes actually reached and lost, rising aspirations require no loss: they turn part of every real gain into felt stagnation.

Absolute and relative purchasing power can move in opposite directions. An individual whose real income rises while that of others rises faster gains purchasing power and loses standing, $\Delta y_i > 0$ while $\Delta r_i < 0$, where r_i is the individual's percentile rank; if well-being depends on income relative to others (Luttmer, 2005; Clark et al., 2008), the second can outweigh the first. Brazilian growth was compressive: between 1995 and 2025, the ratio of income at the 90th percentile to the median fell by about a third (Figure 4), so that for the upper-middle class absolute gains came with a loss of relative standing. The mechanism is bounded, however, because the poorest, whose relative position improved, report losses at least as often as the rich: in November 2025, 75% of respondents earning up to two minimum wages said their purchasing power was lower than a year before, against 67% of those earning more than five (Genial/Quaest, 2025).

5.4 The information environment

Polls on purchasing power ask about the last twelve months, and economic news tracks changes in economic indicators rather than their levels (Soroka et al., 2015), so a

thirty-year gain is invisible to both. Across countries, most consumers also learn about prices from local sources, such as their own shopping, utility bills and acquaintances, which bias their inflation expectations upward (D'Acunto and Weber, 2026). Horizon alone, however, cannot explain the 2025 answers, which diverge from measured income over the same twelve months.

Economic news may also lean negative. In the United Kingdom, media coverage and public opinion respond more strongly to negative than to positive economic information (Soroka, 2006); in Germany, negative unemployment reports outnumbered positive ones by more than the unemployment trend warranted and received more prominent placement (Garz, 2014); and audiences in 17 countries show stronger physiological reactions to negative than to positive news, a demand-side reason to supply it (Soroka et al., 2019). More intensive news coverage makes consumers' inflation expectations more accurate, but not when its tone is negative (Lamla and Lein, 2014). The evidence is mixed: in US newspapers, the asymmetric coverage of good and bad unemployment news is explained by the asymmetric behavior of unemployment itself (Gambetti et al., 2023), and I know of no study of the tone of Brazilian coverage of purchasing power. If price increases make news and stable or rising real incomes do not, households that follow the news receive mostly negative signals. The same selection may affect how the perception itself is gauged: polls that record losses may be more widely reported than those that record gains, so that the perception may seem more widespread than it is.

Assessments of the economy also depend on political allegiance. During Reagan's two terms, inflation fell from 13.5% to 4.1%, yet more than half of strong Democrats said it had worsened (Bartels, 2002), and partisan gaps in US economic expectations have widened without matching differences in spending, which suggests that part of the answers is expressive (Mian et al., 2023). In Brazil, asked about the period since 2023, 42% of respondents said their purchasing power had fallen and 37% that it had risen, most of them attributing the change to the federal government (BTG Pactual and Nexus, 2026). Part of the reported loss may express a political evaluation rather than an economic experience.

Beliefs also spread through social networks: friends' house-price experiences shift individuals' housing beliefs and purchases (Bailey et al., 2018), and inflation expectations co-move across socially connected US counties (García-Lembergman et al., 2023). Under

naive learning, however, a network averages initial beliefs and does not by itself create a common bias (Golub and Jackson, 2010), so social transmission is more plausibly an amplifier of the mechanisms above than an independent source of the gap.

6 Conclusion

The distribution of purchasing power in Brazil improved after the Real. Between 1995 and 2025, real household per capita income grew at every percentile, faster at the bottom than at the top, under the general IPCA and under each of its nine groups; the real minimum wage tripled and inequality fell. The improvement was concentrated in time: in 1995–2003 and 2014–2022, sixteen of the thirty years, real income stagnated across most of the distribution, and nearly all of the gain accrued in the remaining fourteen. The data do not support a loss of purchasing power since the Real; they do show long stretches in which it failed to rise.

Part of the perception of loss can be explained by the structure of the data: percentiles are not people, Brazilian incomes are volatile enough that individual losses may be common over twelve months, and households with atypical baskets or heavy debt service fare worse than the index. The rest lies in perception, and four mechanisms fit the Brazilian facts most closely. The periods of loss weigh more than their length, through loss aversion, reference points set at the previous peak and beliefs that outlast the bad years, and most at the bottom, where the swings were largest. Relative prices moved against groceries and bills, which frequency-weighted and loss-averse perceptions magnify. A substantial part of the gains at the bottom arrived through the minimum wage and transfers, which households may not credit as compensation for higher prices. And the income households consider necessary rises with the income they have. Relative position and the short horizon of polls are bounded by the evidence, while nominal memory, negative news, partisanship and social transmission are plausible amplifiers whose weight remains to be measured.

These mechanisms are hypotheses, and Brazilian data can quantify them. The PNAD Contínua panel identifies the share of people who lost real income in each year, to be set against the share who report a loss; IPCA items and POF purchase records allow a perceived-inflation index weighted by purchase frequency; and a content analysis of news

on prices and incomes can measure how negative the information households receive is. The distinction matters: where losses are real, they can be measured; where prices are misperceived, information can correct them; where households misjudge what prices and incomes mean for them, it cannot, and the perception of lost purchasing power may persist whatever the data show.

References

Albina, I., Bracco, J., Gasparini, L., and Laguinge, L. (2025). Slowing down, heating up: Economic deceleration and social discontent in Latin America. Working Paper 0349, CEDLAS, Universidad Nacional de La Plata.

Bai, J. (1997). Estimation of a change point in multiple regression models. *Review of Economics and Statistics*, 79(4):551–563.

Bai, J. and Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1):47–78.

Bai, J. and Perron, P. (2003). Computation and analysis of multiple structural change models. *Journal of Applied Econometrics*, 18(1):1–22.

Bailey, M., Cao, R., Kuchler, T., and Stroebel, J. (2018). The economic effects of social networks: Evidence from the housing market. *Journal of Political Economy*, 126(6):2224–2276.

Bartels, L. M. (2002). Beyond the running tally: Partisan bias in political perceptions. *Political Behavior*, 24(2):117–150.

Baumann, A., Caprari, L., Kocharkov, G., and Kouvavas, O. (2025). Are real incomes increasing or not? Household perceptions and their role for consumption. *ECB Economic Bulletin*. Issue 1/2025, box.

Bolhuis, M. A., Cramer, J. N. L., Schulz, K. O., and Summers, L. H. (2024). The cost of money is part of the cost of living: New evidence on the consumer sentiment anomaly. Working Paper 32163, National Bureau of Economic Research.

Boyce, C. J., Wood, A. M., Banks, J., Clark, A. E., and Brown, G. D. A. (2013). Money, well-being, and loss aversion: Does an income loss have a greater effect on well-being than an equivalent income gain? *Psychological Science*, 24(12):2557–2562.

Brachinger, H. W. (2008). A new index of perceived inflation: Assumptions, method, and application to Germany. *Journal of Economic Psychology*, 29(4):433–457.

BTG Pactual and Nexus (2026). País se divide em percepção de mudança no poder de compra do brasileiro. `https://www.nexus.fsb.com.br/estudos-divulgados/pais-se-divide-em-percepcao-de-mudanca-no-poder-de-compra-do-brasileiro-aponta-btg-nexus/`. Survey of 11–13 Sep. 2026, $n = 2{,}003$. Accessed on 12 Oct. 2026.

Carroll, C. D. (2003). Macroeconomic expectations of households and professional forecasters. *Quarterly Journal of Economics*, 118(1):269–298.

Clark, A. E., Frijters, P., and Shields, M. A. (2008). Relative income, happiness, and utility: An explanation for the Easterlin paradox and other puzzles. *Journal of Economic Literature*, 46(1):95–144.

Cummings, R., Harris, B., and Mahoney, N. (2024). The paradox between the macroeconomy and household sentiment. Brookings Institution, `https://www.brookings.edu/articles/the-paradox-between-the-macroeconomy-and-household-sentiment/`.

D'Acunto, F., Malmendier, U., Ospina, J., and Weber, M. (2021). Exposure to grocery prices and inflation expectations. *Journal of Political Economy*, 129(5):1615–1639.

D'Acunto, F. and Weber, M. (2026). Information and macroeconomic expectations: Global evidence. Working Paper 35511, National Bureau of Economic Research.

Duesenberry, J. S. (1949). *Income, Saving, and the Theory of Consumer Behavior*. Harvard University Press, Cambridge, MA.

Easterlin, R. A. (1995). Will raising the incomes of all increase the happiness of all? *Journal of Economic Behavior & Organization*, 27(1):35–47.

Engbom, N. and Moser, C. (2022). Earnings inequality and the minimum wage: Evidence from Brazil. *American Economic Review*, 112(12):3803–3847.

Ferreira, F. H. G., Leite, P. G., and Litchfield, J. A. (2008). The rise and fall of Brazilian inequality: 1981–2004. *Macroeconomic Dynamics*, 12(S2):199–230.

Gambetti, L., Maffei-Faccioli, N., and Zoi, S. (2023). Bad news, good news: Coverage and response asymmetries. Finance and Economics Discussion Series 2023-001, Board of Governors of the Federal Reserve System.

García-Lembergman, E., Hajdini, I., Leer, J., Pedemonte, M., and Schoenle, R. (2023). The expectations of others. Working Paper 23-22, Federal Reserve Bank of Cleveland.

Garz, M. (2014). Good news and bad news: Evidence of media bias in unemployment reports. *Public Choice*, 161(3):499–515.

Genial/Quaest (2024). 63% dos brasileiros acham que poder de compra hoje é menor do que há um ano. CNN Brasil, `https://www.cnnbrasil.com.br/economia/macroeconomia/quaest-63-dos-brasileiros-acham-que-poder-de-compra-hoje-e-menor-do-que-ha-um-ano/`. Accessed on 12 Oct. 2026.

Genial/Quaest (2025). Para 72%, poder de compra está menor que há 1 ano. CNN Brasil, `https://www.cnnbrasil.com.br/economia/money/macroeconomia/genial-quaest-para-72-poder-de-compra-esta-menor-que-ha-1-ano/`. Accessed on 30 Sep. 2026.

Georganas, S., Healy, P. J., and Li, N. (2014). Frequency bias in consumers' perceptions of inflation: An experimental study. *European Economic Review*, 67:144–158.

Giambiagi, F., Villela, A., Castro, L. B., and Hermann, J., editors (2016). *Economia Brasileira Contemporânea: 1945–2015*. Elsevier, Rio de Janeiro, 3rd edition.

Goedhart, T., Halberstadt, V., Kapteyn, A., and van Praag, B. M. S. (1977). The poverty line: Concept and measurement. *Journal of Human Resources*, 12:503–520.

Golub, B. and Jackson, M. O. (2010). Naïve learning in social networks and the wisdom of crowds. *American Economic Journal: Microeconomics*, 2(1):112–149.

Graham, C. and Pettinato, S. (2002). Frustrated achievers: Winners, losers and subjective well-being in new market economies. *Journal of Development Studies*, 38(4):100–140.

Grimm, M. (2007). Removing the anonymity axiom in assessing pro-poor growth. *Journal of Economic Inequality*, 5(2):179–197.

Guerreiro, J., Hazell, J., Lian, C., and Patterson, C. (2026). Why do workers dislike inflation? Wage erosion and conflict costs. *Econometrica*, 94(5):1651–1684.

Hacıoğlu Hoke, S., Feler, L., Mitchell, S., and Chylak, J. (2025). Tracking consumer sentiment versus how consumers are doing based on verified retail purchases. *FEDS Notes*.

Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384.

IBGE (2019). POF 2017-2018: Famílias com até R$ 1,9 mil destinam 61,2% de seus gastos à alimentação e habitação. Agência IBGE Notícias, `https://agenciadenoticias.ibge.gov.br/agencia-sala-de-imprensa/2013-agencia-de-noticias/releases/25598-pof-2017-2018-familias-com-ate-r-1-9-mil-destinam-61-2-de-seus-gastos-a-alimentacao-e-habitacao`. Accessed on 30 Sep. 2026.

IBGE (2020). 30% da população com menores rendimentos vivem com menos que o necessário. Agência IBGE Notícias, `https://agenciadenoticias.ibge.gov.br/agencia-noticias/2012-agencia-de-noticias/noticias/29491-30-da-populacao-com-menores-rendimentos-vivem-com-menos-que-o-necessario`. Accessed on 12 Oct. 2026.

IBGE (2026). Extended national consumer price index – IPCA. `https://www.ibge.gov.br/en/statistics/economic/prices-and-costs/17129-extended-national-consumer-price-index.html`. Accessed on 30 Sep. 2026.

Jaravel, X. (2021). Inflation inequality: Measurement, causes, and policy implications. *Annual Review of Economics*, 13:599–629.

Jenkins, S. P. and Van Kerm, P. (2006). Trends in income inequality, pro-poor income growth, and income mobility. *Oxford Economic Papers*, 58(3):531–548.

Jonung, L. (1981). Perceived and expected rates of inflation in Sweden. *American Economic Review*, 71(5):961–968.

Kahneman, D. and Tversky, A. (1979). Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291.

Kőszegi, B. and Rabin, M. (2006). A model of reference-dependent preferences. *Quarterly Journal of Economics*, 121(4):1133–1165.

Kuchler, T. and Zafar, B. (2019). Personal experiences and expectations about aggregate outcomes. *Journal of Finance*, 74(5):2491–2542.

Lamla, M. J. and Lein, S. M. (2014). The role of media for consumers' inflation expectation formation. *Journal of Economic Behavior & Organization*, 106:62–77.

Luttmer, E. F. P. (2005). Neighbors as negatives: Relative earnings and well-being. *Quarterly Journal of Economics*, 120(3):963–1002.

Malmendier, U. and Nagel, S. (2016). Learning from inflation experiences. *Quarterly Journal of Economics*, 131(1):53–87.

Medeiros, M., Souza, P. H. G. F., and Castro, F. A. (2015). O topo da distribuição de renda no Brasil: Primeiras estimativas com dados tributários e comparação com pesquisas domiciliares (2006–2012). *Dados – Revista de Ciências Sociais*, 58(1):7–36.

Mian, A., Sufi, A., and Khoshkhou, N. (2023). Partisan bias, economic expectations, and household spending. *Review of Economics and Statistics*, 105(3):493–510.

Morgan, M. (2017). Extreme and persistent inequality: New evidence for Brazil combining national accounts, surveys and fiscal data, 2001–2015. WID.world Working Paper 2017/12, World Inequality Lab.

Prais, S. J. (1959). Whose cost of living? *Review of Economic Studies*, 26(2):126–134.

Ravallion, M. and Chen, S. (2003). Measuring pro-poor growth. *Economics Letters*, 78(1):93–99.

Shafir, E., Diamond, P., and Tversky, A. (1997). Money illusion. *Quarterly Journal of Economics*, 112(2):341–374.

Shiller, R. J. (1997). Why do people dislike inflation? In Romer, C. D. and Romer, D. H., editors, *Reducing Inflation: Motivation and Strategy*, pages 13–70. University of Chicago Press, Chicago.

Soares, S., Osório, R. G., Soares, F. V., Medeiros, M., and Zepeda, E. (2009). Conditional cash transfers in Brazil, Chile and Mexico: Impacts upon inequality. *Estudios Económicos*, pages 207–224. Número extraordinario.

Soroka, S., Fournier, P., and Nir, L. (2019). Cross-national evidence of a negativity bias in psychophysiological reactions to news. *Proceedings of the National Academy of Sciences*, 116(38):18888–18892.

Soroka, S. N. (2006). Good news and bad news: Asymmetric responses to economic information. *Journal of Politics*, 68(2):372–385.

Soroka, S. N., Stecula, D. A., and Wlezien, C. (2015). It's (change in) the (future) economy, stupid: Economic indicators, the media, and public opinion. *American Journal of Political Science*, 59(2):457–474.

Souza, P. H. G. F., Gonçalves, S., Portella, A., Osorio, R. G., and Firpo, S. (2026). Income volatility and poverty dynamics in Brazil. *Economía LACEA Journal*, 25(1):40–55.

Souza, P. H. G. F., Osorio, R. G., Paiva, L. H., and Soares, S. (2019). Os efeitos do Programa Bolsa Família sobre a pobreza e a desigualdade: Um balanço dos primeiros quinze anos. Texto para Discussão 2499, Instituto de Pesquisa Econômica Aplicada (Ipea), Brasília.

Stantcheva, S. (2024). Why do we dislike inflation? *Brookings Papers on Economic Activity*, 55(1):1–46.

Teixeira, F. O., Oliveira, I. C. L. d., and Ferreira, P. G. C. (2017). Consumers' inflation expectations in Brazil. *Estudos Econômicos*, 47(3):509–529.

Weber, M., D'Acunto, F., Gorodnichenko, Y., and Coibion, O. (2022). The subjective inflation expectations of households and firms: Measurement, determinants, and implications. *Journal of Economic Perspectives*, 36(3):157–184.

A Additional figures

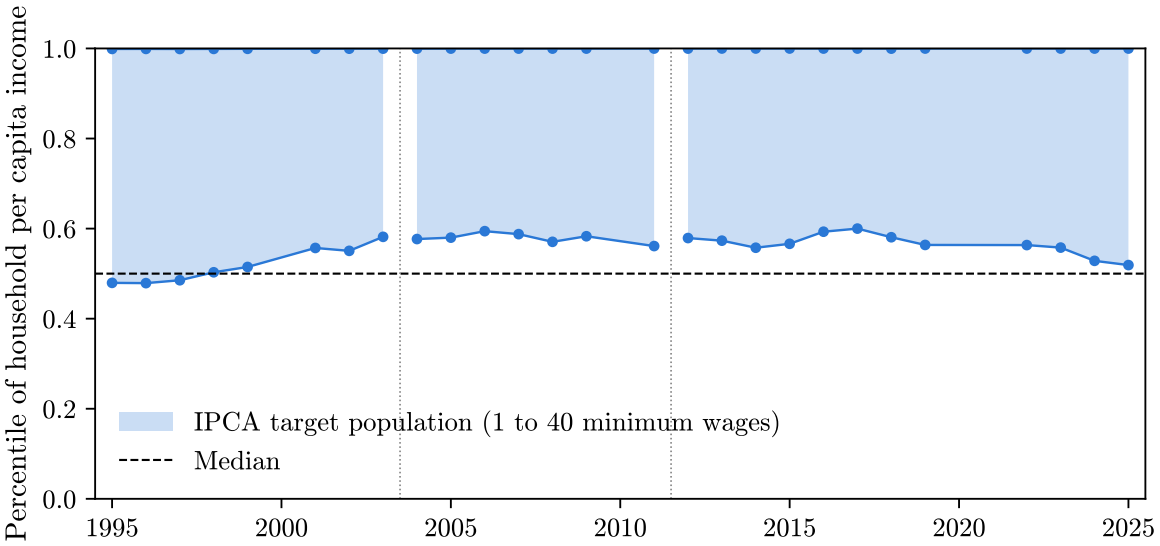


Figure A1: Percentiles of household per capita income – one to forty minimum wages

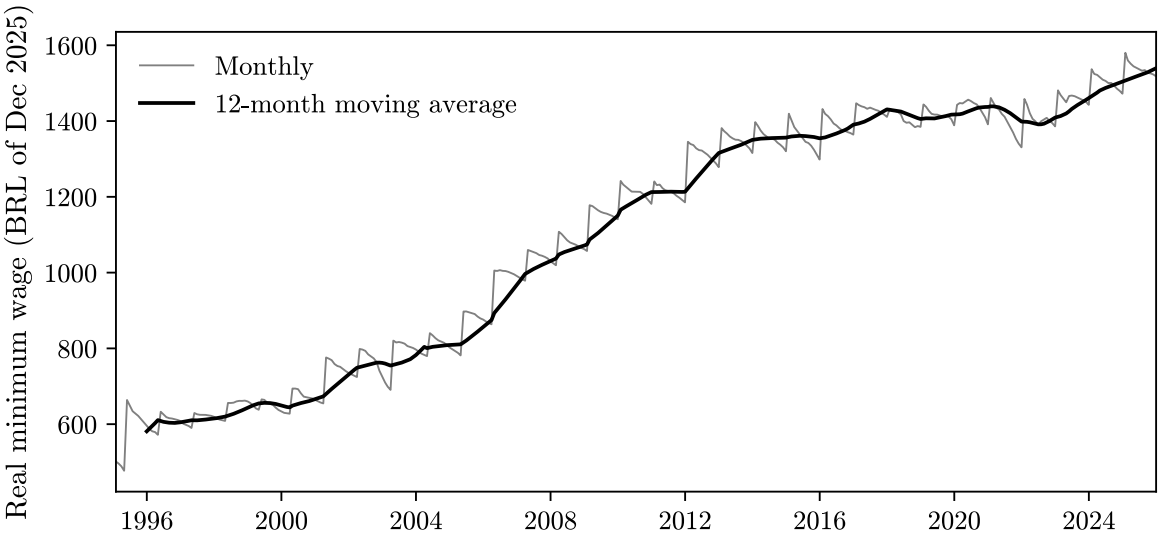


Figure A2: Real minimum wage

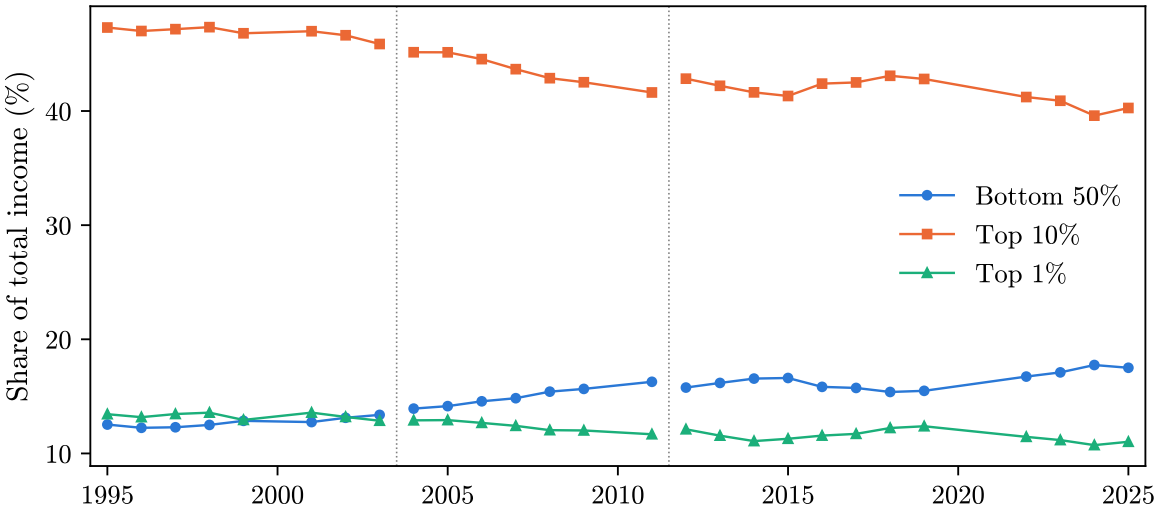


Figure A3: Income shares of the bottom 50%, the top 10% and the top 1%

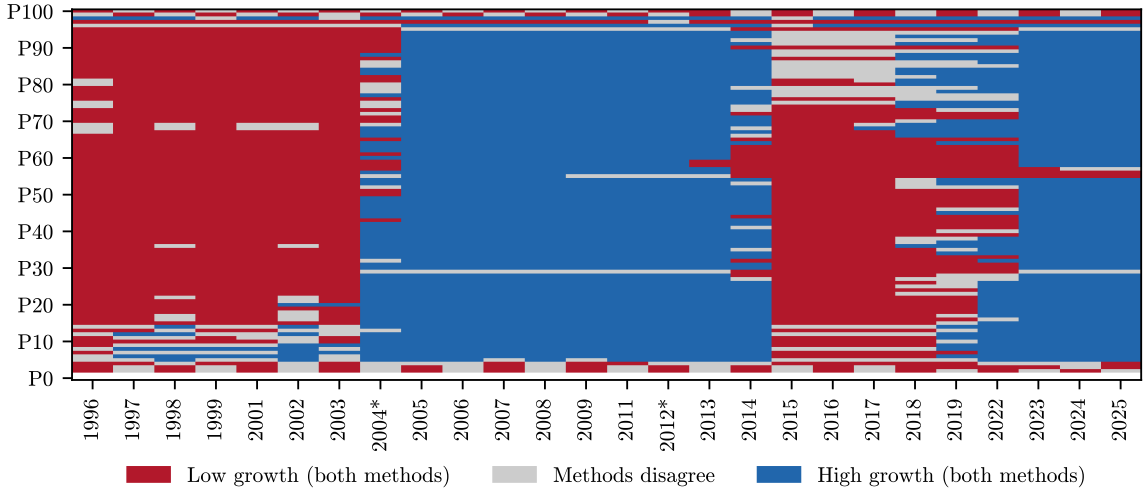


Figure A4: Regimes of real growth by percentile: consensus of the two methods

Note: Each survey at each percentile is classified by the Bai–Perron and Markov-switching methods (Section 3.1). Grey: the methods disagree; white: not classified (zero incomes). * As in Figure 6.

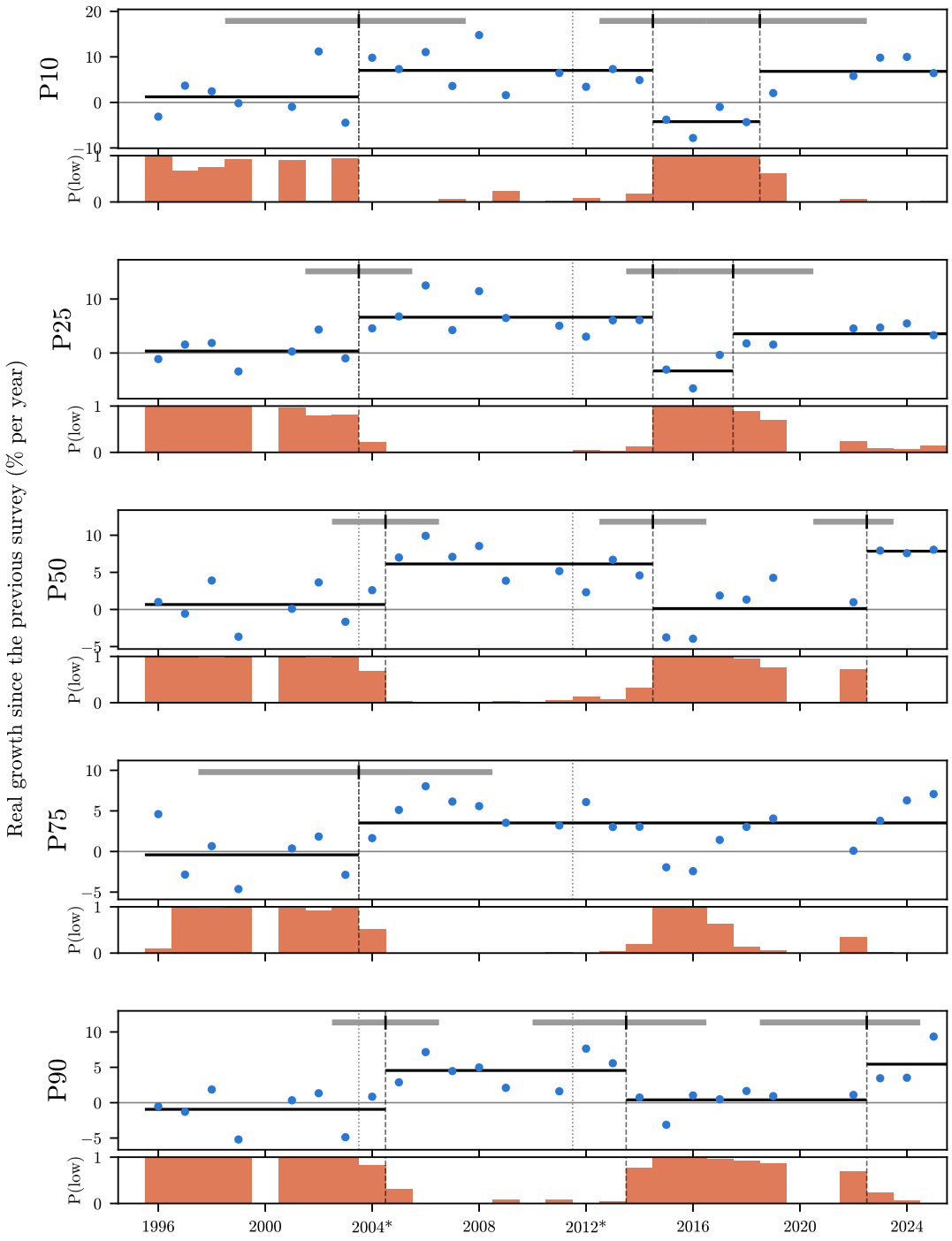


Figure A5: Bai–Perron breaks and Markov low-growth probability at five percentiles

Note: Upper panels: real growth since the previous survey (dots), Bai–Perron regime means (lines) and 90% confidence intervals of the break dates (grey bars). Lower panels: smoothed probability of the Markov low-growth regime. * As in Figure 6.